

AI-BOX Edge Box Management Platform — User Manual (English)

Version: 1.2 | Document Type: User Guide | Audience: End Users, Operations Staff

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1. Product Introduction

1.1 What is AI-BOX?

AI-BOX is an **Edge computing box management platform** for deploying, running, and managing AI video analysis tasks on edge devices. Using a visual drag-and-drop interface, you can combine input sources (e.g., RTSP cameras), algorithms (e.g., pedestrian detection, face recognition), and outputs (e.g., network streaming, alerts, recording) into complete analysis pipelines.

1.2 Core Features

Module	Description
Task Management	Create, edit, start, stop, copy, delete tasks; group management
Flow Editing	Drag-and-drop canvas, configure node parameters, connect data flows
Video Preview	Multi-channel live preview, multiple layouts, full screen
Alert Notification	View, acknowledge, delete alerts; alert popup toggle
Face Library	Manage face lists (whitelist/blacklist/VIP, etc.)
Recording	Browse, preview, delete recordings; space cleanup
System Settings	Device info, version, upgrade, reboot, change password

2. System Requirements & Access

2.1 Browser Requirements

- **Recommended:** Chrome 90+, Edge 90+, Safari 14+
- JavaScript must be enabled
- HTTPS recommended (required for recording preview)

2.2 Access URL

- **Local:** `https://<device-IP>:port` Or `https://<device-IP>:port/aibox/`
- **Example:** `https://192.168.1.100:8099/aibox/`

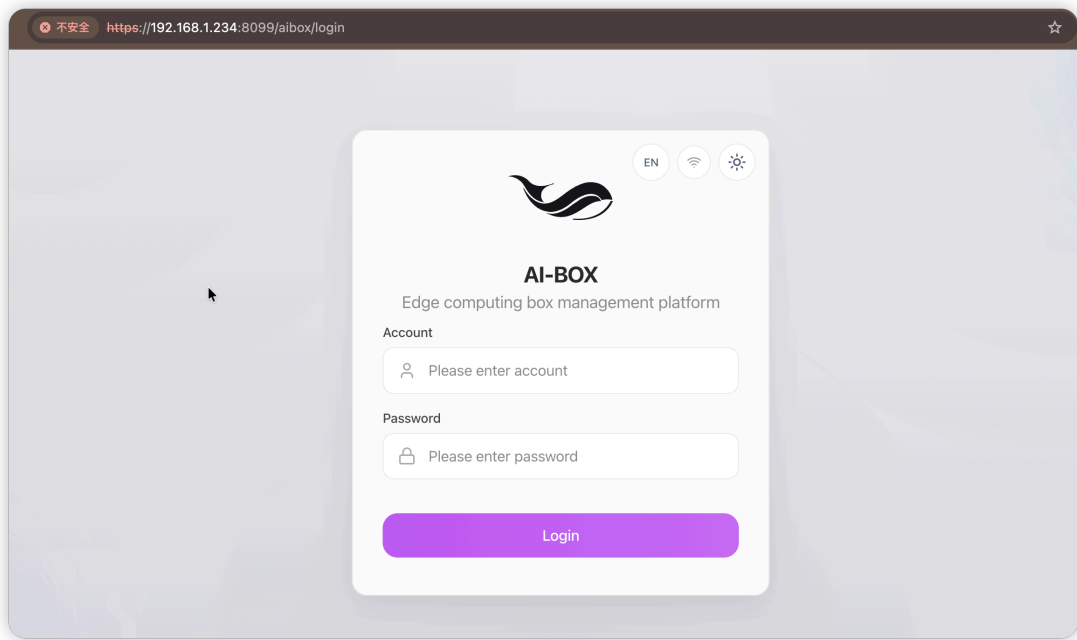
2.3 Device Connection

- The web app connects to the device via WebRTC. Ensure the **AI-BOX app is running** on the device.
- If connection fails, check: device online status, network connectivity, firewall rules.

3. Quick Start: Login

3.1 Open Login Page

1. Enter the access URL in the browser address bar (see 2.2)



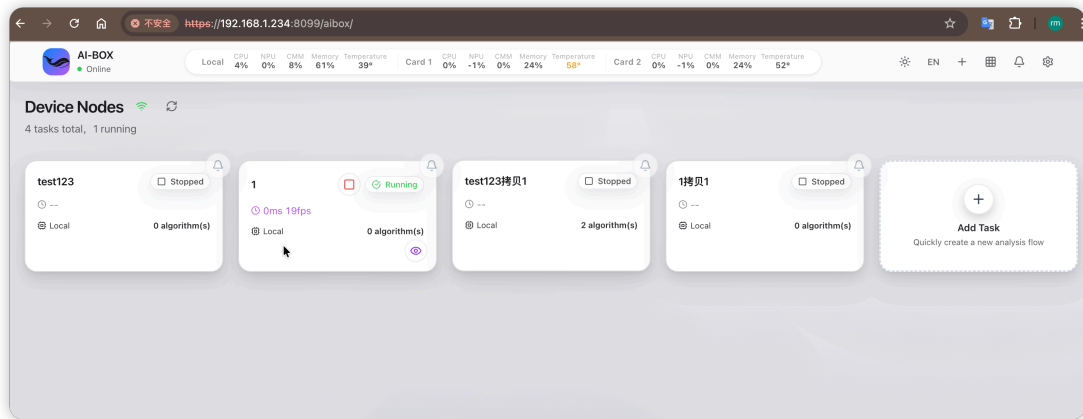
2. If not logged in, you will be redirected to the login page.

3.2 Login Steps

Step	Action	Notes
1	Enter username in the Account field	Default is usually <code>admin</code> for local deployment
2	Enter password in the Password field	Input is masked; default is <code>admin</code>
3	Click Login	The system first connects via WebRTC, then validates credentials; either step failing will show an error
4	Wait for connection	Button shows "Logging in..."; do not click repeatedly; usually completes in seconds
5	Success	Redirects to home (Device Nodes); session conflict may occur if already logged in elsewhere

3.3 Login Result

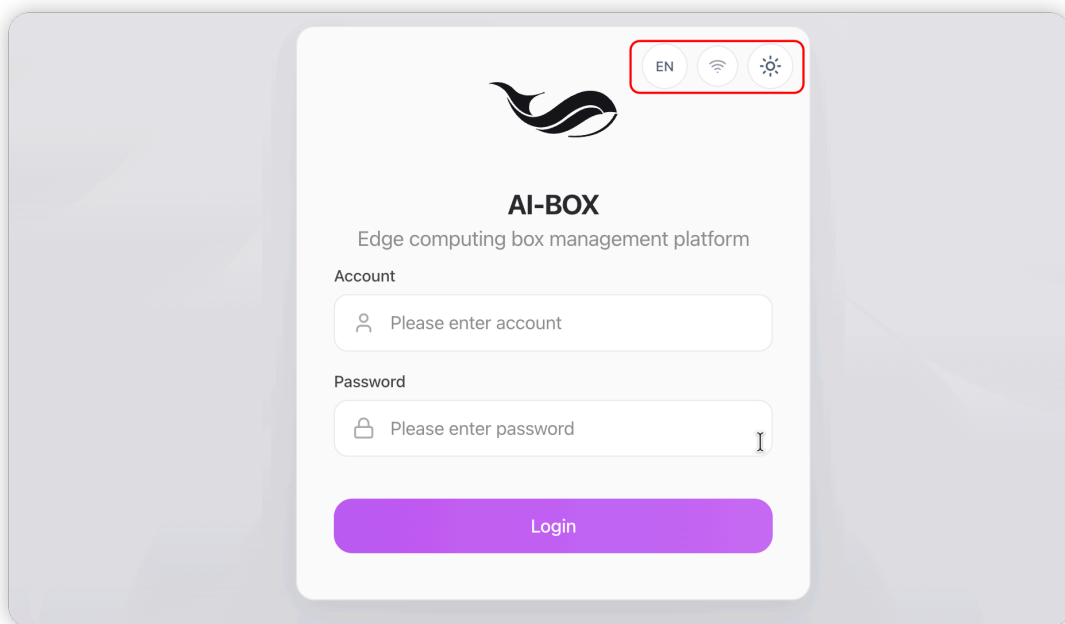
- **Success:** Redirects to home (Device Nodes)



- **Failure:** Error message at top or in popup (e.g., "Device connection failed", "Login expired")
- If **device connection failed:** Confirm the device app is running and network is reachable, then refresh and retry.

3.4 Theme & Language (Login Page)

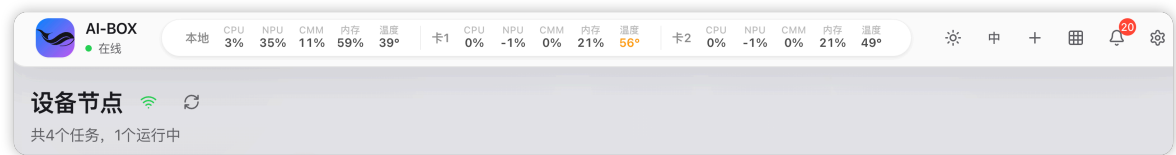
- **Theme:** Toggle light/dark via sun/moon icon on the login page.
- **Language:** Switch Chinese/English from the main Settings menu after login.



4. Main Interface Overview

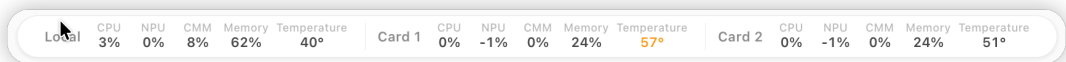
4.1 Top Bar (Header)

Area	Content
Left	Logo, device name, online/offline status; shows "Reconnect" when offline
Center	Device status capsules: CPU, NPU, memory, temperature (double-click for details)
Right	Theme, language, add task, multi-view, alert bell, settings menu



4.2 Device Status Capsules

- Shows real-time resource usage for **local** and **compute cards**
- **Colors:** Green = normal, Orange = warning, Red = critical
- **Double-click** to open device details popup

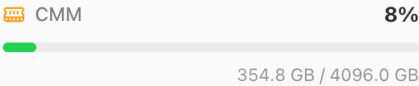
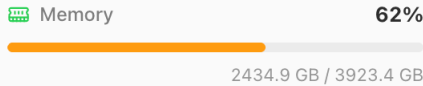
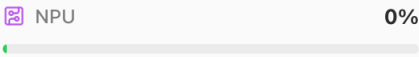
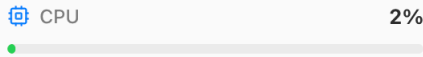


Device Details



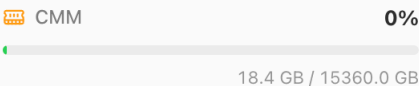
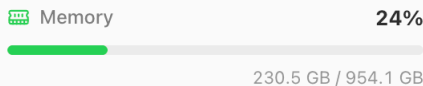
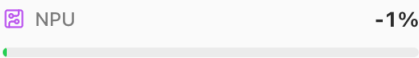
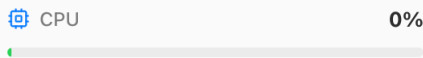
Local

40°C



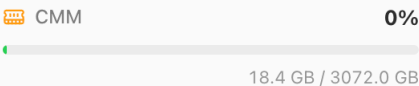
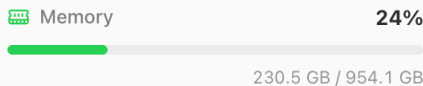
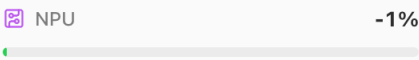
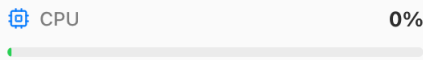
Card 1

57°C



Card 2

50°C



Storage



System Info

App Version unknown

Firmware Version Ax_Version V3.6.2

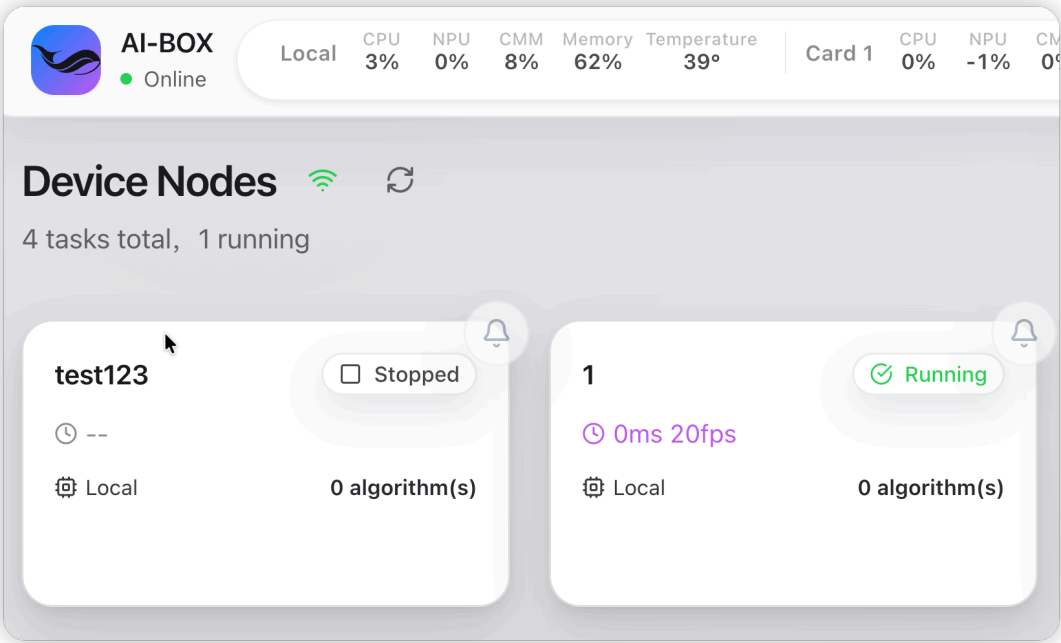
Device GUID ax650n_1821ec5e9a29264d8824f4fbdc1737c0

Authorization 已授权

System Time 2026-02-24 17:05:17:778

4.3 Main Content Area

- **Title:** "Device Nodes", with "X tasks total, X running"
- **Connection icon:** Green = connected, Orange = connecting, Red = disconnected
- **Refresh:** When connected, refreshes data without disconnecting; when disconnected, attempts to reconnect
- **Task cards & groups:** See next section



4.4 When Device Not Connected

A red banner appears with:

- Error description
- **Reconnect:** Try to re-establish connection
- **Re-login:** Clear session and return to login page

5. Task Management

5.1 Task Card Types

Type	Appearance	Description
Group Card	Folder icon + count	Holds multiple tasks; click to expand/collapse
Task Card	Name, status, algorithm count	Single analysis task
Add Task Card	Dashed border + plus	Quick create new task

5.2 Create New Task

Method 1: Click "Add Task" card

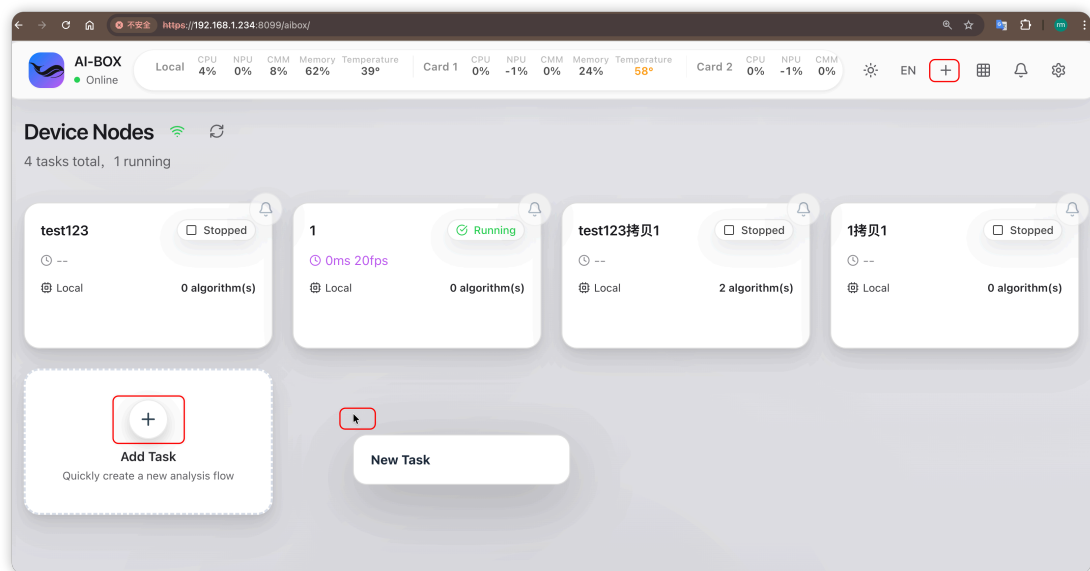
1. Find the "Add Task" card (dashed border, "+" in center)
2. Click it
3. Enter task name in the dialog
4. Click "Create and Edit" to open the flow editor

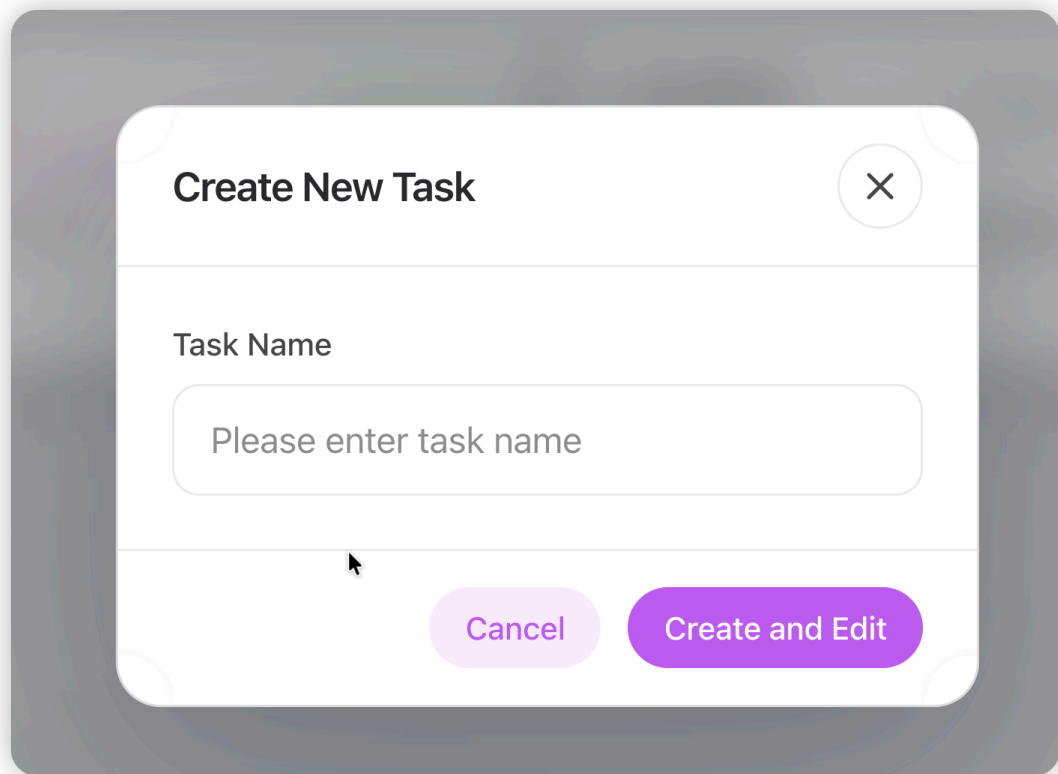
Method 2: Click top "+" button

1. Click the plus icon in the top bar
2. Same steps as above

Method 3: Right-click on blank area

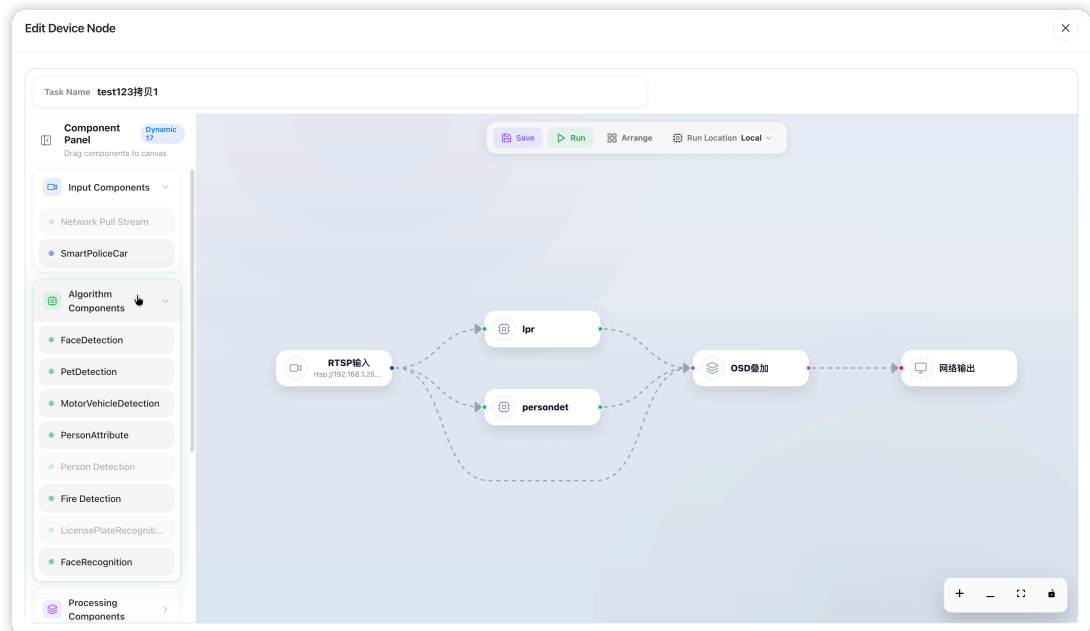
1. Right-click on any blank area (not on cards or buttons)
2. Choose "New Task"
3. Enter name in the dialog





5.3 Edit Task

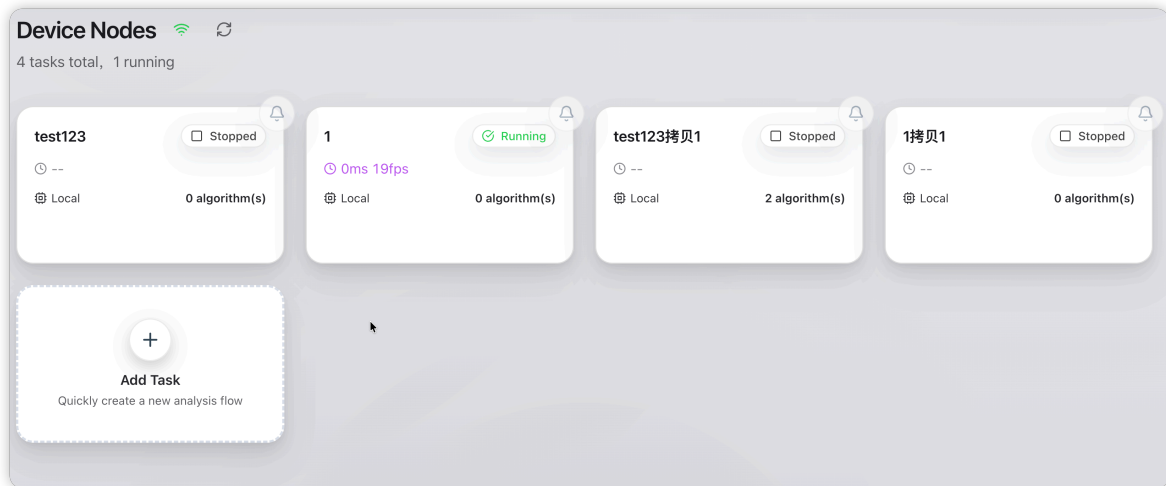
1. **Double-click** the task card, or click the **Edit** icon on the card
2. Opens the flow editor (see Chapter 6)



5.4 Start & Stop Task

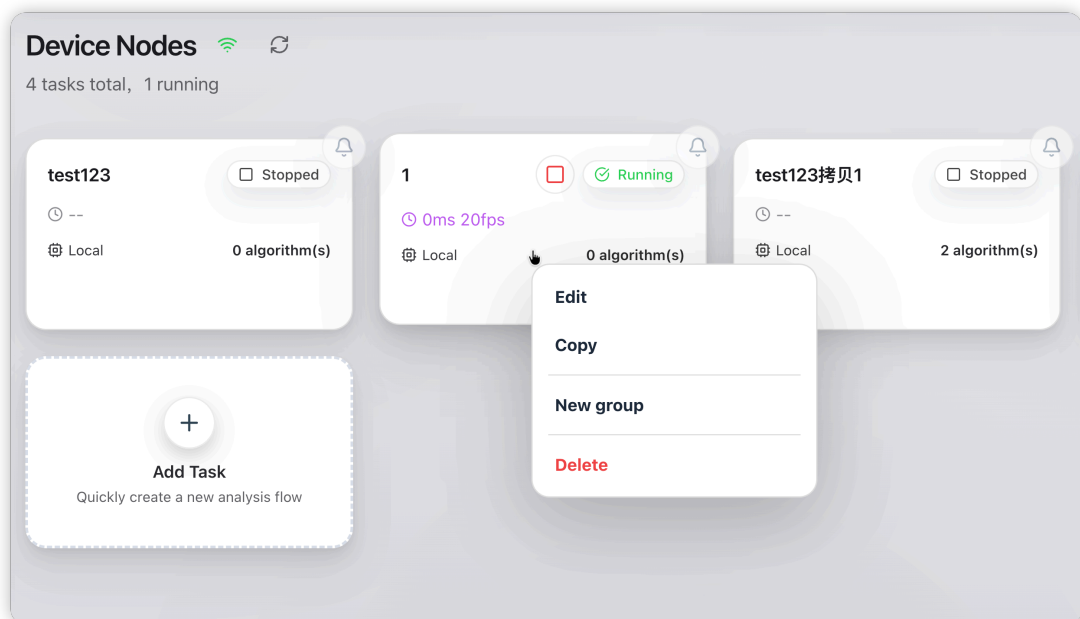
- **Start:** Click **play icon** (▶) on the task card
- **Stop:** Click **stop icon** (■) on the task card
- **Status:** Running (green), Stopped (gray), Error (red), Schedule Paused (purple)

⚠ Cannot start/stop when device is not connected.



5.5 Copy Task

1. Right-click the task card
2. Choose "Copy"
3. A copy is created with "Copy" suffix in the name



5.6 Delete Task

1. Right-click the task card
2. Choose "Delete" (red item)
3. Confirm in the dialog
4. ⚠ Deletion is irreversible.

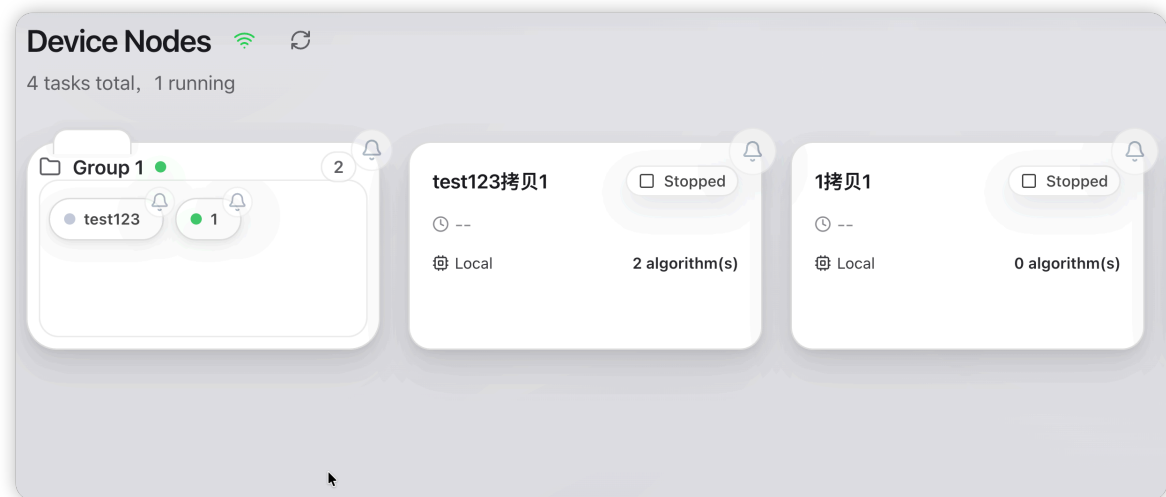
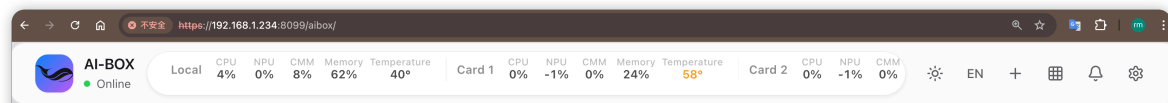
5.7 Group Management

Create group: Right-click task card → "New Group"; or drag a task onto a group card

Move task: Right-click task → "Move to X"; or drag task onto group card

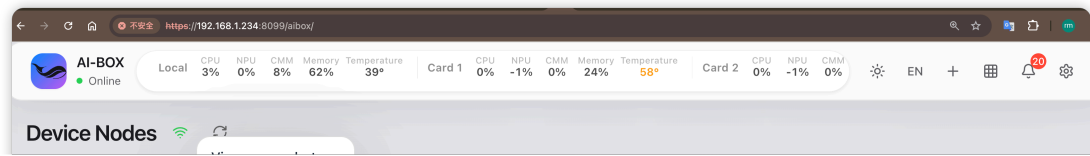
Rename group: Right-click group card → "Rename"

Remove from group: Right-click task → "Remove from Group" (when task is in a group)



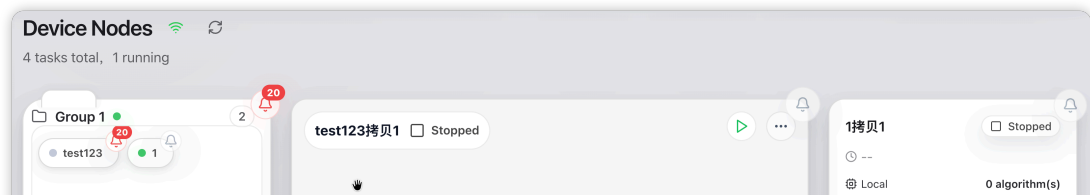
5.8 Alert Bell on Task Card

- Shows unread alert count for that task/group
- Click to open alert panel filtered by that task



5.9 Task Card Expand/Collapse

- **Single click:** Expands to show flow preview (nodes and connections)
- **Click blank area again:** Collapses to small card
- **Double-click:** Opens flow editor directly
- When expanded: **Preview**, **Edit** buttons at bottom-right; hover for **Start/Stop**, **More**



5.10 Drag to Copy (Option/Alt + Drag)

- Hold **Option** (Mac) or **Alt** (Windows) and **drag** task card to another group or blank area
- Releases a copy at the target; no need to right-click Copy first

6. Flow Editor

6.1 Open Flow Editor

- Double-click task card, or click Edit icon on the card

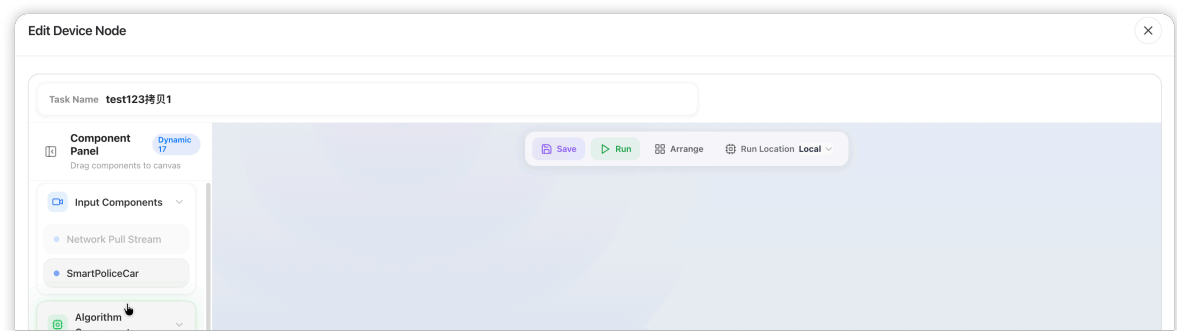
- New tasks: click "Create and Edit" after creation

6.2 Editor Layout

Area	Description
Left panel	Input, Algorithm, Processing, Output components; collapsible
Center canvas	Drag, connect, layout area
Top toolbar	Save, Run, Arrange, Close
Run location	Dynamic or specific compute card

6.3 Add Nodes to Canvas

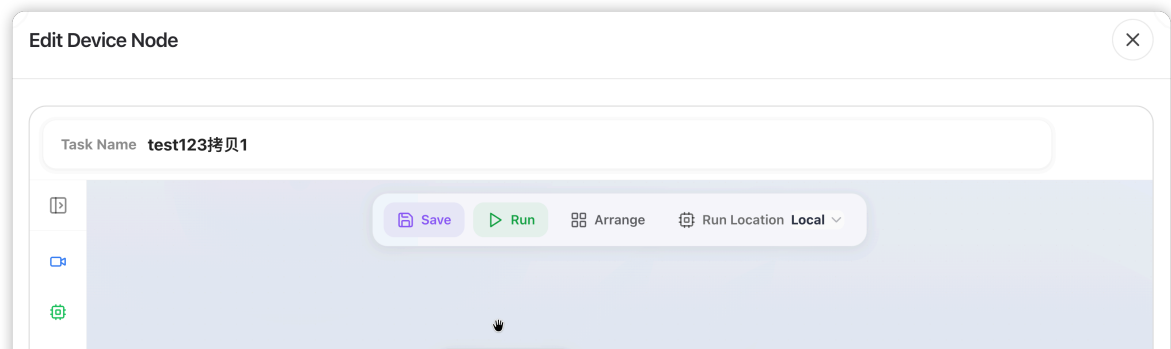
1. Locate the component in the left panel (Input, Algorithm, Processing, Output)
2. Drag it onto the canvas
3. If a duplicate type exists, you'll see "Task already contains xxx"
4. Release; node appears at drop location



6.4 Connect Nodes

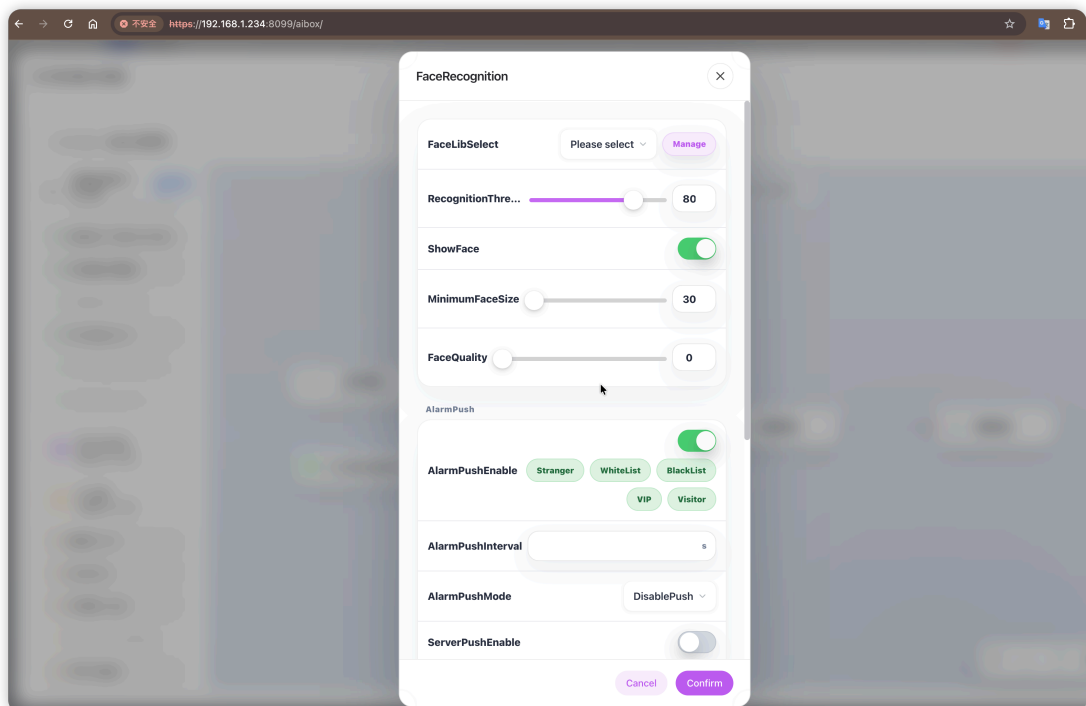
1. Each node has left (input/target) and right (output/source) handles
2. Drag from **source left handle** to **target right handle**
3. Canvas auto-arranges after new connections
4. Use top "Arrange" button to re-layout

⚠ Direction must be: Input → Algorithm → Processing → Output.



6.5 Configure Node Parameters

1. **Double-click** the node on the canvas
2. Fill the form in the config popup
3. Use "Test Connection", "ONVIF Search" etc. if available
4. Click "OK"; config is applied locally; click "Save" or "Run" at top to sync to device



6.6 Canvas Operations

- **Pan:** Drag on blank canvas
- **Zoom:** Mouse wheel
- **Delete node:** Select node, press Delete

- **Delete edge:** Select edge, press Delete
- **Arrange:** Click top "Arrange" button

6.7 Save & Run

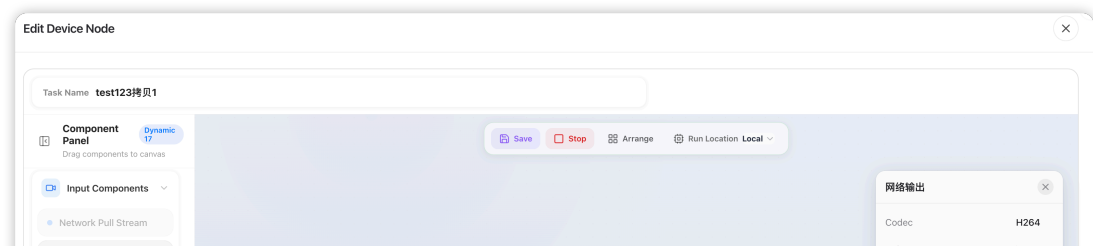
- **Save:** Top "Save" button
- **Run:** Top "Run" button (saves then starts)
- If task references deleted plugins, save/run will fail; remove or replace those nodes first

6.8 Canvas Context Menu

- Right-click **node**: Edit node, Delete node
- Right-click **edge**: Delete edge
- Click outside to close menu

6.9 Run Location

- Top "Run location" dropdown: **Dynamic, Local, Compute Card 1, Compute Card 2**, etc.
- **Dynamic:** Device assigns automatically; specific choice fixes to that unit



7. Node Configuration

Node config is driven by **formList** from the device/plugins. Common control types:

7.1 Input Controls

Type	Description
input	Text or number; placeholder, unit
textarea	Multi-line text
inputNumber	Min/max range
password	Masked input

7.2 Selection Controls

Type	Description
select	Dropdown choices
switch	Toggle on/off
slider	Drag to set value

7.3 Read-only & Status

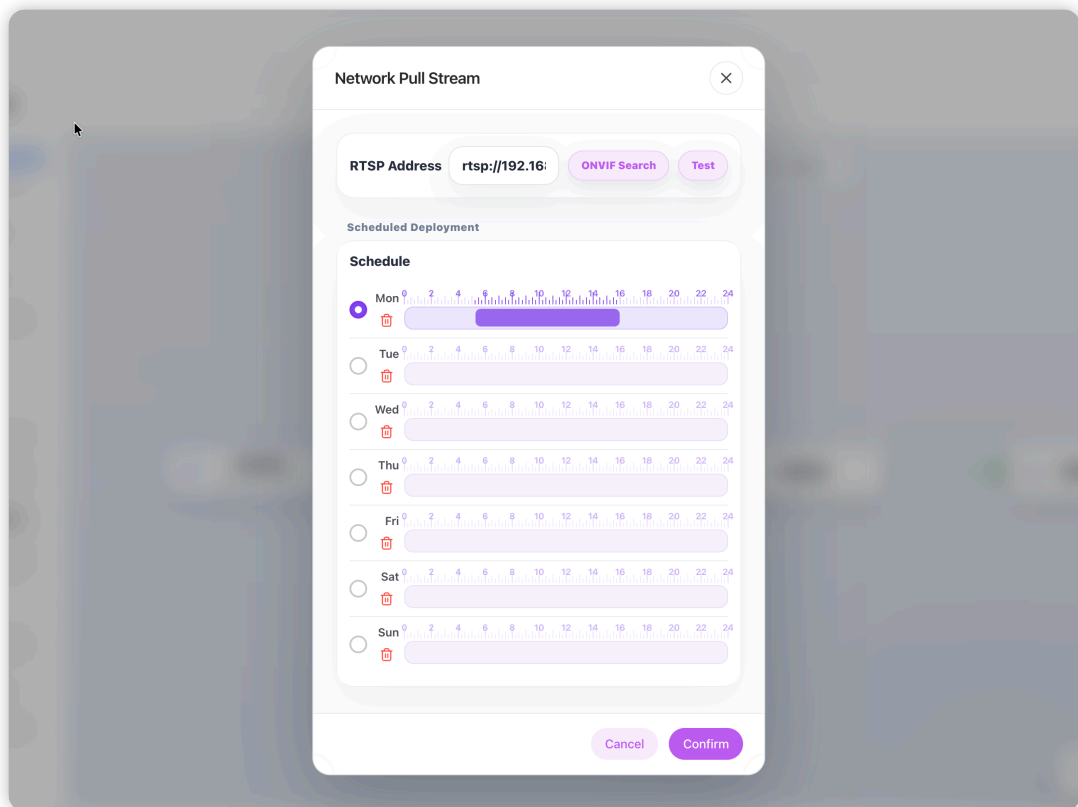
Type	Description
readOnly	Display only
status	Status label with color (OK/RUNNING=green, FAILED=red, WARNING=orange)

7.4 Special Config

Type	Description
button	Execute action (e.g., Test Connection, ONVIF Search)
divider	Section header
schedule	Time-based schedule (Monday–Sunday, 24h)
regionDraw	Draw detection regions on snapshot

7.5 Schedule (Time-based Control)

1. In RTSP Input node config, open "Schedule"
2. Set enabled times by day, 15-minute slots
3. Enable/disable days
4. Copy to other days, clear day, etc.
5. Task runs only in enabled periods



7.6 Region Drawing (Detection Area)

1. Task must be **saved** and device **connected**
2. In algorithm node config, click "Draw Detection Region"
3. Click "Refresh Snapshot" to get current frame

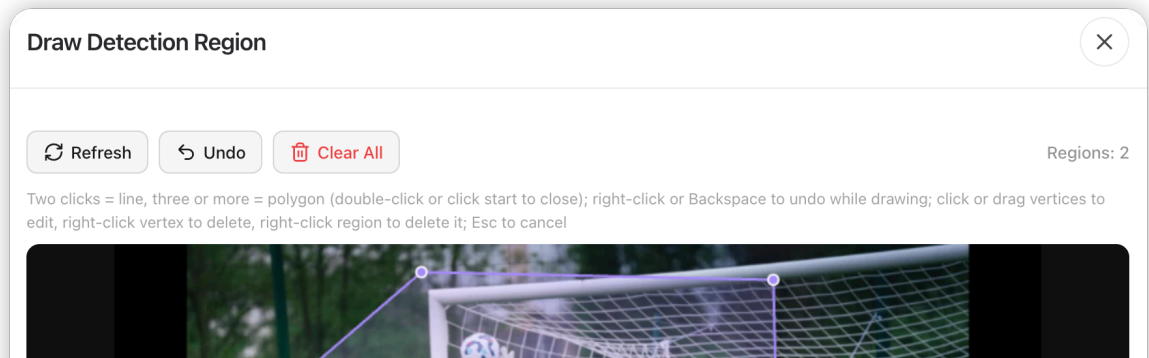
Drawing:

Type	Action
Line	Two clicks to complete, or double-click
Polygon	Click to add vertices (3+); double-click or click start to close
Undo last step	Right-click or Backspace while drawing
Cancel	Esc

Editing & deletion:

Action	Description
Drag vertex	Left-click and drag vertex to reshape
Delete vertex	Left-click vertex to select, press Delete ; or right-click vertex
Delete region	Right-click inside region or on segment; choose "Delete this region"; overlapping targets are highlighted in red
Select region	Right-click region; click elsewhere to close menu or deselect

Use Undo, Clear All as needed; supports multiple regions; save to apply.



7.7 Quick Buttons

Button	Description
Test Connection / RTSP Test	Validate RTSP URL
ONVIF Search	Search ONVIF devices on LAN
Face Library	Open face library (face recognition nodes)
Select Recording Path	Choose recording storage path

8. Runtime Node Status

When a task runs, nodes and edges show live status.

8.1 Edge Status

Status	Color	Meaning
idle	Gray dashed	Not running or unknown
ok	Green dashed (animated)	Data flowing normally
stuck	Red dashed	Blocked or error
paused	Purple dashed	Schedule paused (outside enabled time)

8.2 Node Hover (detail_info)

- **Hover** over a node in a running task to see a popup
- Shows **detail_info** from device: format, resolution, fps, latency, etc.
- Click values to copy to clipboard

8.3 Form Status in Config

- Some read-only/status fields in node config are updated live via **form_status**

8.4 Node Timeout

- If a node has no valid data for ~10 seconds, task health becomes **warning** or **error**
- Task card shows "Run Warning" or "Run Error"; edges may turn red

9. Task Card Status

9.1 Task & Health Status

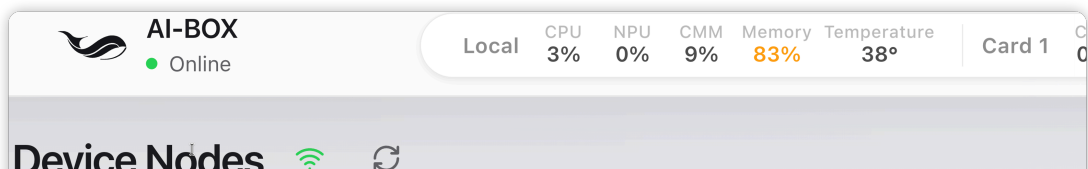
Status	Icon/Color	Meaning
Running	Green check	Normal
Stopped	Gray square	Stopped
Starting/Stopping	Orange loading	In progress
Error	Red X	Critical error
Run Warning	Orange triangle	Some nodes timed out
Schedule Paused	Purple pause	Within schedule, outside enabled time

9.2 Latency & FPS

- When **running** and has **Network Output** node: shows **Xms Yfps** or **Xms** or --

9.3 Schedule Resume Countdown

- When schedule-paused, card shows countdown like "3d 2h 15m 30s until resume"

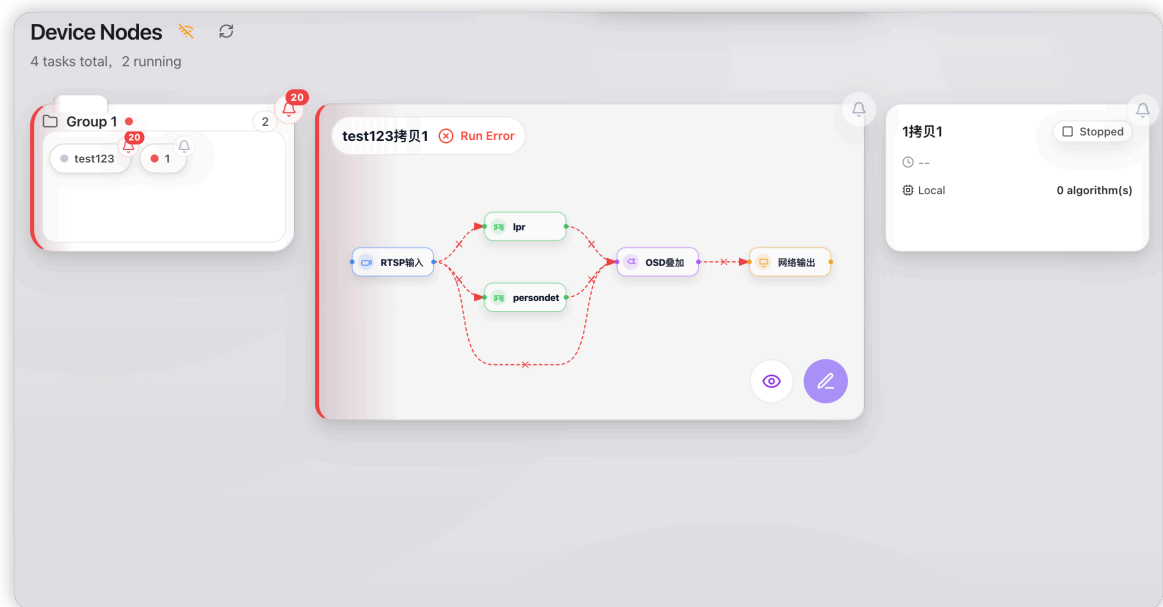


9.4 Run Location & Algorithm Count

- **Run location:** Local, Compute Card 1, etc.
- **X algorithm(s):** Number of algorithm nodes

9.5 Card Left Color Bar

Color	Meaning
Purple	Schedule paused
Red	Run error
Orange	Run warning



10. Node Connection Guidelines

10.1 Data Flow Order

- Standard: **Input** → **Algorithm** → **Processing** → **Output**
- Input nodes as start; output nodes as end
- Algorithm/processing can be parallel or serial

10.2 Connection Rules

- Connect source **left** to target **right**
- No duplicate node types in one task (unless plugin supports multi-input)

10.3 Tips

1. Build flow first, then configure parameters
2. Preview requires **Network Output** node
3. Alerts require **Alert Output** node
4. Run will auto-save if there are unsaved changes

10.4 Common Errors

Symptom	Cause	Fix
Cannot connect	Wrong direction	Connect source right → target left
No preview	No Network Output	Add and connect Network Output
No alerts	No Alert Output	Add and connect Alert Output
Node grayed	Plugin removed	Remove or replace node

11. Video Preview & Multi-view

11.1 Open Multi-view

- Click **grid icon** in top bar
- Or **drag** task to preview dock at bottom

11.2 Add Preview Source

- **Drag** task to preview dock
- Or select from multi-view dialog

11.3 Layout

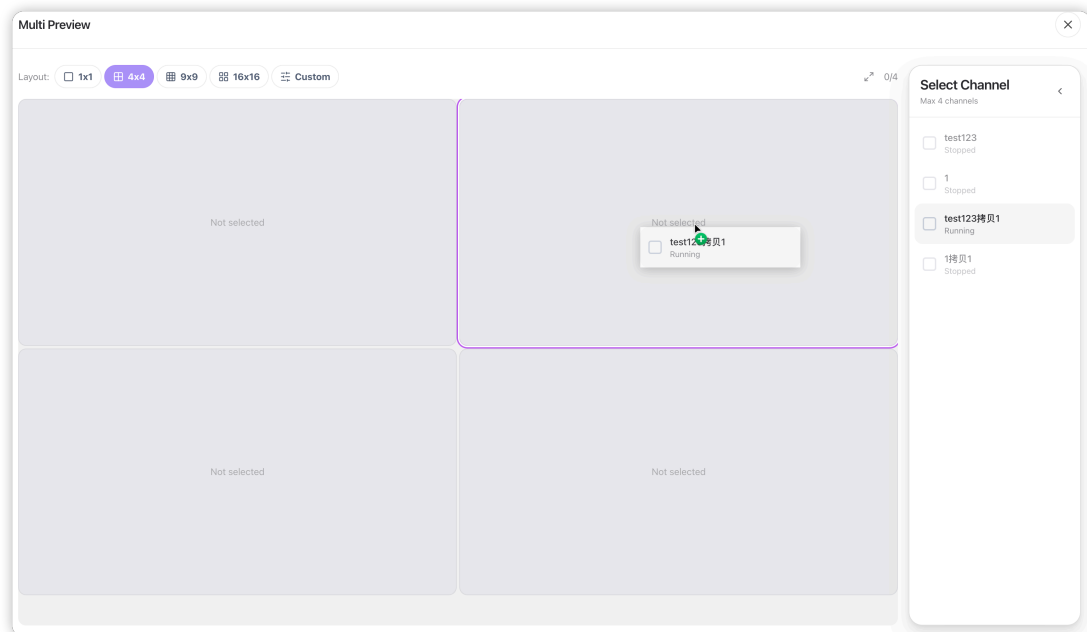
- Auto, 2/4/9/16 grid, custom

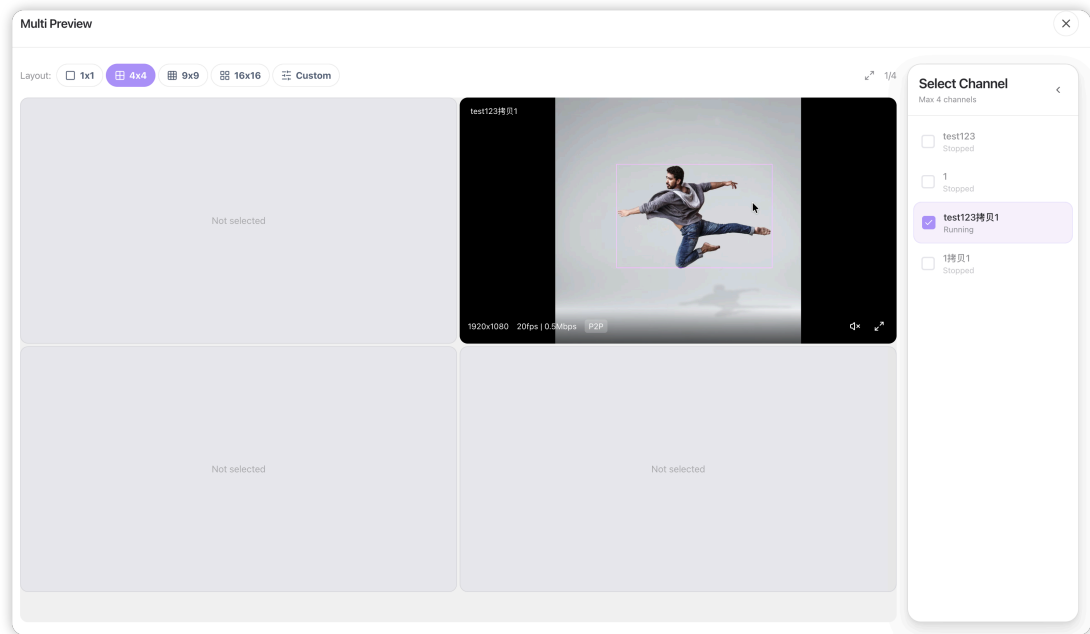
11.4 Full Screen & Close

- Full screen button; Close All

11.5 Notes

- Task must have **Network Output** node
- Must **start task** before preview





12. Alert Management

12.1 Open Alert Panel

- Click **bell icon** in top bar; red number = unread count

12.2 Alert List

Action	Description
View	Click to see details, large image
Acknowledge	Mark as read
Mark All Read	Mark all as read
Delete	Single or batch (with confirm)
Load More	Pull up at bottom

12.3 Alert Popup

- Settings menu → "Alert Popup" toggle

12.4 Filter

- By task, by alert type



13. Plugin Management

This chapter is intended for customers and operations teams. It explains how to upload, install, enable, disable, uninstall, replace, and roll back plugins from the web UI.

13.1 Entry and Status Overview

Where to open it:

- From the top menu: **Plugin Manager**
- In some builds, the same entry is also available from the settings menu

Common status meanings in the plugin list:

Status	Meaning	Recommended user action
Loaded	The plugin has already been loaded successfully by the running service	Safe to use in flows and tasks
Installed	The package is already stored on the device, but the current service has not fully loaded it yet	Usually restart the service and verify again
Disabled	The plugin was manually turned off	Do not use it in new tasks until re-enabled
Pending disable	The plugin is still referenced by tasks and cannot be fully disabled yet	Stop the related tasks and check status again
Pending update	The new package is already prepared, but the old version is still waiting for task references to be released	Perform the change during a low-traffic window
Error	Loading, initialization, or validation failed	Check package integrity, compatibility, and runtime dependencies

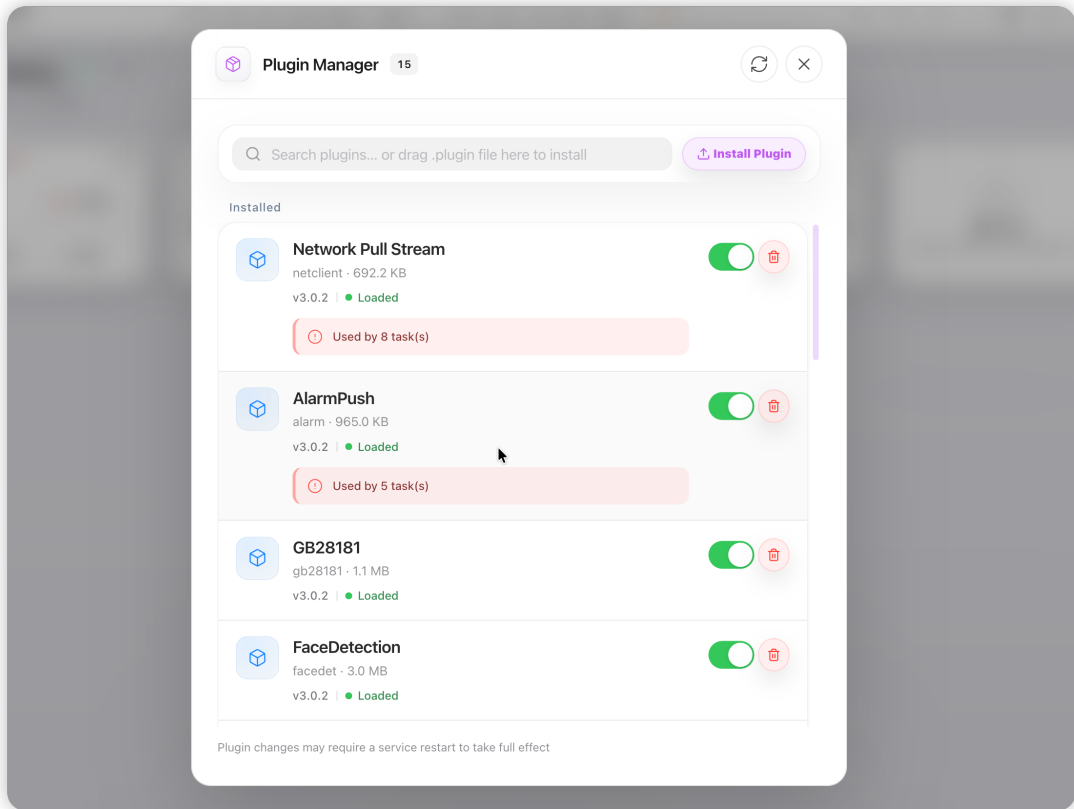
Compatibility warning:

- If the UI shows a compatibility warning, the platform declared by the plugin does not fully match the current device architecture, OS, or chip.
- You may validate it in a test environment first. Confirm compatibility before using it in actual tasks.

13.2 Uploading and Installing Plugins

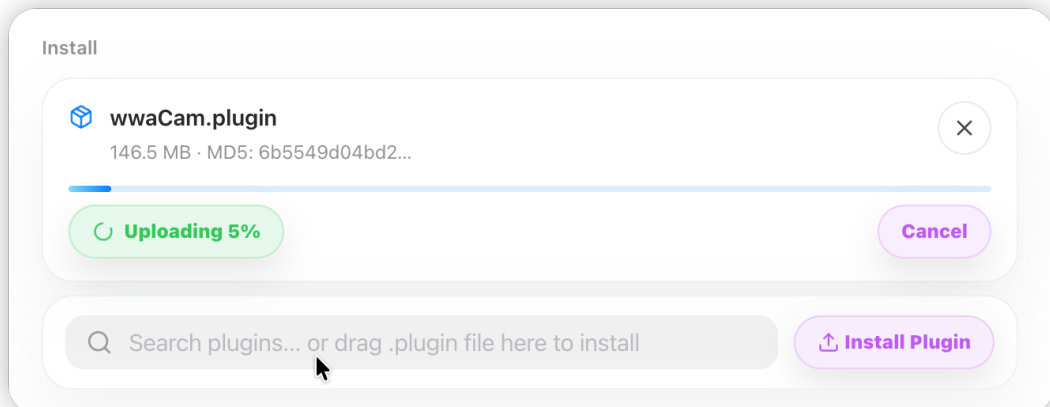
13.2.1 Standard installation procedure

1. Open **Plugin Manager**.
2. Click **Install Plugin**, or drag a `.plugin` file into the upload area.
3. Wait for upload, validation, and package inspection to finish.
4. Confirm plugin name, type, version, and compatibility information.
5. Click **Install**.
6. After installation succeeds, follow the UI prompt to **Restart Service**.
7. After the page reconnects, verify that the plugin state becomes **Loaded**.



13.2.2 What users may see during installation

- Small plugin packages usually complete validation and installation quickly.
- Large plugin packages may take noticeably longer than normal configuration changes.
- Even after the upload finishes, the page may still show a processing state while the device continues the installation. This is often normal.



13.2.3 What to check before installing

1. Make sure the file is a valid `.plugin` package.
2. Confirm the plugin type matches your planned use, such as `llm.plugin`, `record.plugin`, or `persondet.plugin`.
3. Confirm the version matches the intended release plan.
4. Check whether the current device architecture, OS version, and chip are compatible.

5. Confirm device storage is sufficient so installation does not fail after upload.

13.3 Replacement, Upgrade, and Same-type Plugin Override

If a plugin of the same type already exists on the device, installing a new package of that type is typically treated as a replacement or upgrade.

13.3.1 Recommended workflow

1. Confirm which tasks are currently using that plugin.
2. Stop the related tasks during a low-traffic window.
3. Install the new plugin package.
4. Restart the service after installation.
5. Verify plugin version, state, and related task behavior after restart.

13.3.2 Typical system behavior

- If the old plugin is still referenced by running tasks, direct replacement may be blocked until those tasks are released.
- After a successful replacement, the system usually preserves the old package as a managed backup for rollback.
- Before restart, the UI may show the plugin as **Installed** or **Pending update** rather than **Loaded**. This is expected transitional behavior.

13.3.3 Operation Tips

- Before replacement, make sure related tasks have been stopped.
- After replacement, return to the task page and confirm tasks still work normally.

13.4 Enabling, Disabling, and When Changes Take Effect

13.4.1 Enabling a plugin

1. Locate the target plugin in the list.
2. Turn on the enable switch.
3. If the UI asks for **Restart Service**, follow that prompt.
4. After restart, return to the plugin list and confirm the state is correct.

13.4.2 Disabling a plugin

1. Turn off the plugin switch.
2. If no task is using the plugin, the system may disable it immediately.
3. If tasks still reference it, the state will become **Pending disable**.
4. Stop the related tasks and, if needed, restart the service before checking the final state again.

13.4.3 Common misunderstandings

- **Installed is not the same as Loaded:** installed only means the package is already stored on disk.
- **Pending disable is not necessarily a failure:** it usually means a task still references the plugin.
- **No immediate state change after enable does not always mean an error:** the system may still be processing the change or waiting for restart.

13.5 Uninstall and Rollback

13.5.1 Uninstalling a plugin

1. Confirm that no task still references the plugin.
2. If tasks still contain nodes from that plugin, remove or replace those nodes and save the tasks first.
3. Stop all related running tasks.
4. Return to **Plugin Manager** and uninstall the plugin.
5. Verify that the plugin disappears from the list or moves to the expected final state.

Notes:

- If the UI reports that the plugin is still in use, do not keep retrying blindly. Remove task dependencies first.
- If old tasks still contain nodes from a removed plugin, those tasks may no longer save or run until the nodes are cleaned up manually.

13.5.2 Rolling back a plugin

1. Rollback is available only if a previous replacement kept an older backup package.
2. Stop every task that still references the plugin before rollback.
3. Trigger rollback and wait for the previous version to be restored.
4. Restart the service after rollback and confirm the plugin state and version again.

Notes:

- If the UI reports that rollback is unavailable, no preserved backup package exists on the device.

13.6 Operation Tips

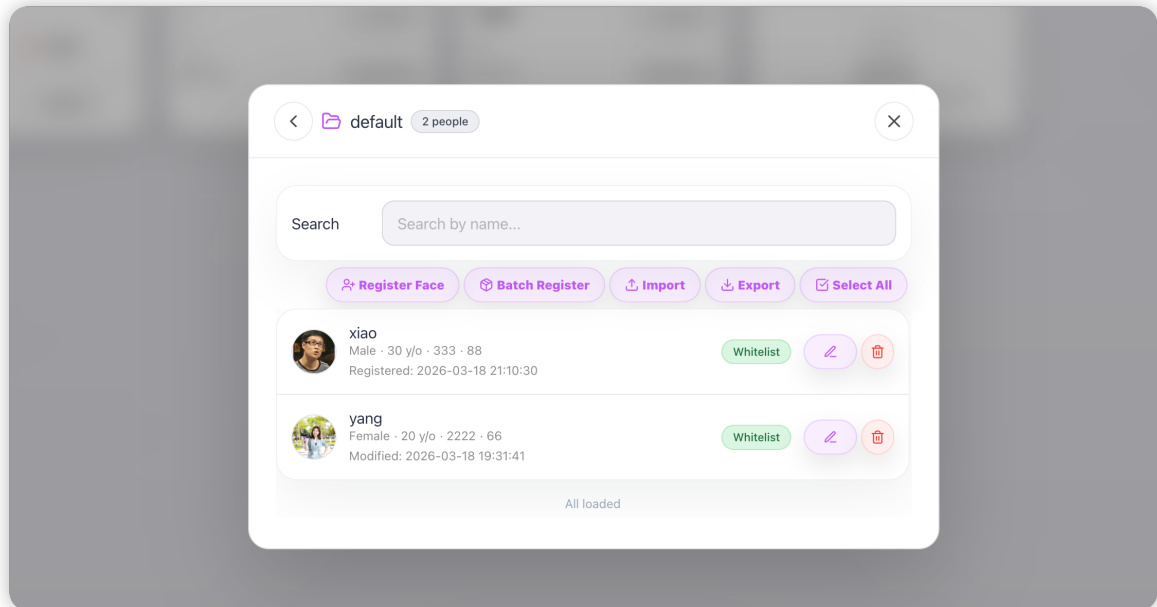
1. During upload, installation, or replacement, do not refresh the page or click buttons repeatedly.
 2. If the UI asks for **Restart Service**, follow the prompt.
 3. If the page disconnects during restart, wait for it to reconnect.
 4. If plugin status stays abnormal for a long time, refresh the page and check again; if needed, confirm the plugin package is correct.
-

14. Plugins: Face Library & Recording

14.1 Face Library

Entry

- Settings → Face Library, if the shortcut is enabled
- Or click **Manage** in the face-recognition node configuration



Interface overview

- The upper level is the **face-library list**, where you can create, rename, and delete libraries
- After entering a library, you can search, register, batch register, import, export, and select faces in batch
- Supported person types are: **Whitelist, Blacklist, VIP, Visitor**

14.1.1 Create and open a face library

1. Click **New Face Library**
2. Enter the library name
3. Confirm to create the library
4. Click the library item to enter the face list

14.1.2 Register a single face

1. Enter the target face library
2. Click **Register Face**
3. Select one clear face photo
4. Enter the name
5. Optionally fill in gender, age, phone number, and person type
6. Optionally fill in extended fields: `PID`, `Work ID`, ID card number, IC card number,

department

7. Click confirm. The system uploads the image first, then extracts face features on the device and stores the record

Recommendations

- Use a **single-person, frontal, clear, unobstructed** face photo whenever possible
- Avoid group photos, strong backlight, tiny faces, or heavily compressed images
- If the system reports "no face detected" or "registration failed", replace the image with a clearer photo and try again

14.1.3 Search, edit, and delete

- The search box performs **fuzzy search by name**
- Click the edit button or double-click a row to update face information
- When changing the photo, the system uploads the new image first and then saves the profile
- Click the delete button to remove one face
- In selection mode, you can run batch delete, batch move, and batch category update

14.1.4 Batch registration

Batch registration is suitable when many people need to be added at once, such as project initialization or periodic library updates.

Steps:

1. Enter the target face library.
2. Click **Batch Register**.
3. Select multiple images or the supported batch-import file.
4. Fill in the required options shown by the page.
5. Click confirm and wait for the system to finish processing.
6. Review the result dialog for success count, failure count, and failure reasons.

Recommendations

- Use clear file names so staff can check imported records easily later.
- For larger imports, work in smaller batches instead of importing everything at once.
- Failed items can usually be retried separately without affecting the successfully imported records.

14.1.5 Export a face library

1. Enter the target face library.
2. Click **Export**.
3. The browser downloads an export file. Save it to the customer PC or backup location.

Notes

- The export file includes the current library records and related images.

- If the library is empty, the system may not generate an export file.
- For larger libraries, export may take more time. Keep the page open until the download completes.

14.1.6 Import a face library

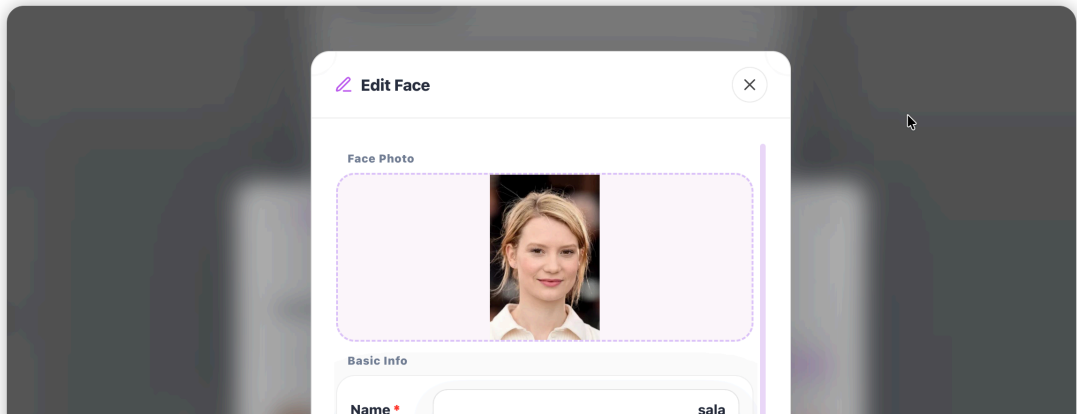
1. Enter the target face library.
2. Click **Import**.
3. Select the exported library file.
4. Choose the duplicate-name handling policy shown by the page, such as skip or overwrite.
5. Click confirm and wait for import to finish.
6. Check the result message for succeeded, skipped, and failed items.

Recommendations

- Make sure you are importing into the correct target library.
- For migration or large updates, export a backup before importing new data.
- If import fails, reduce the batch size and retry in smaller groups.

14.1.7 Recommendations and current limitations

- The current version is suitable for routine maintenance, batch import/export by library, and small to medium deployments
- For large libraries, split operations **by library and by time window**. For commercial deployments, avoid keeping all data in one oversized library whenever possible
- For large exports, use **wired network + recent Chrome/Edge + a PC with sufficient memory**. Keep the page open and avoid network changes during export
- When a single import/export job is very large, the browser may still show long wait time, increased memory usage, or stalled progress. This is a normal boundary of large package handling
- For commercial projects, plan naming rules, library grouping, and unique identifiers such as `PID` or `Work ID` in advance to reduce duplicate data and manual maintenance

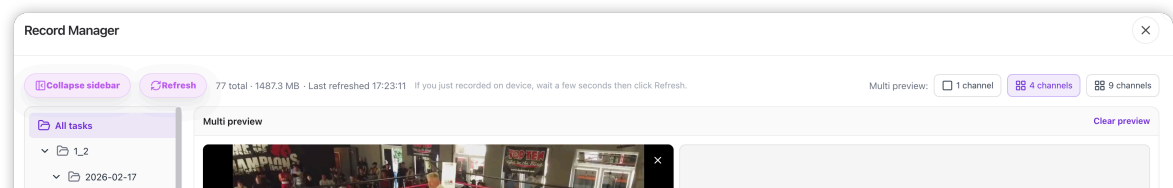


14.2 Recording

Entry: Settings → Recording

Actions: Browse by task/date, preview, delete, space cleanup, refresh

If recording preview is unavailable, first confirm the device recording function is working and the page is connected to the device.



15. LLM Plugin Usage (Review Mode & Detector Mode)

This chapter only explains how customers can use the LLM plugin in the web UI.

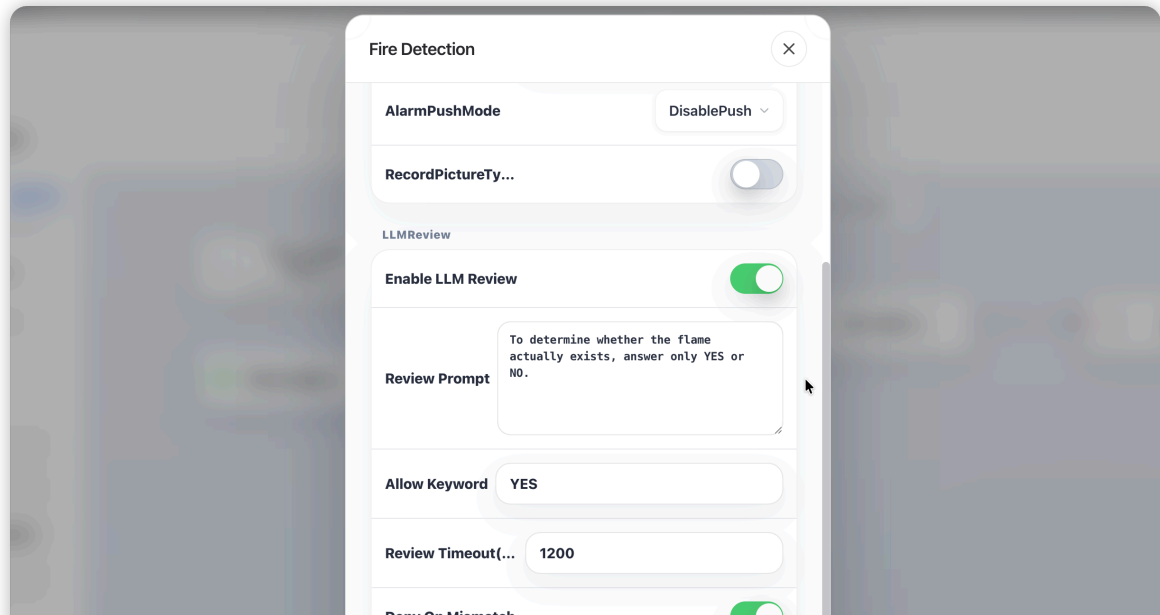
15.1 Function Overview

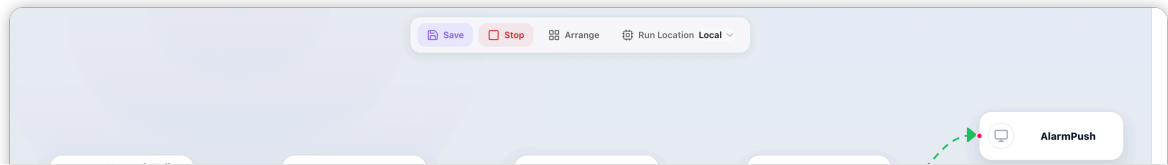
The LLM plugin is commonly used in two ways:

1. **Review Mode:** an existing algorithm detects first, and LLM performs a second check.
2. **Detector Mode:** LLM directly checks whether the target exists in the scene.

15.2 Review Mode Steps

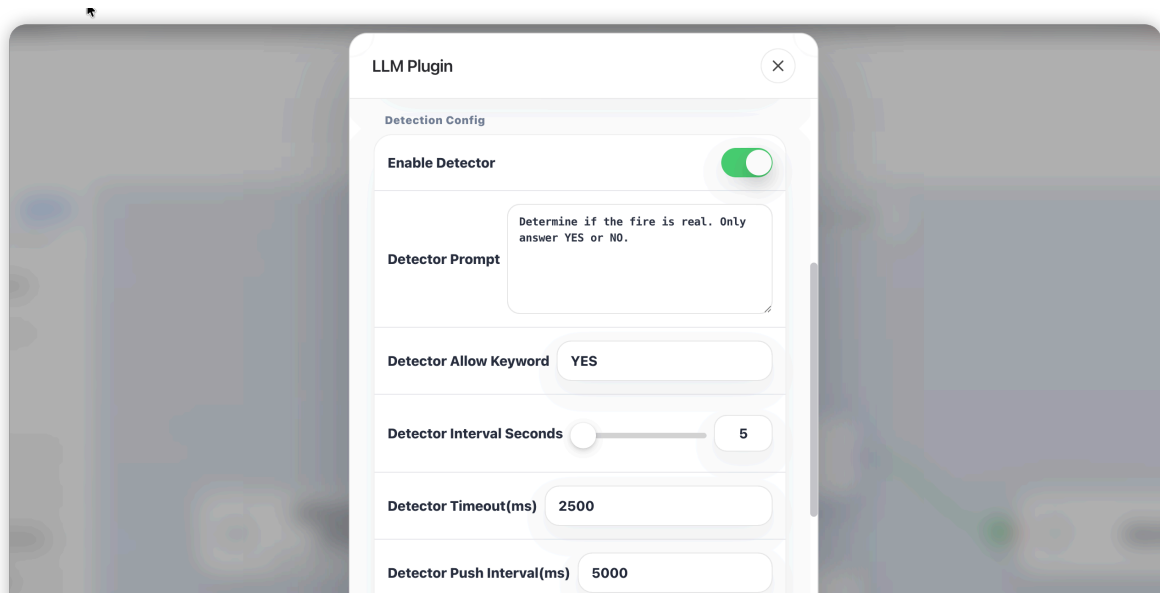
1. Create a task and open the flow editor.
2. Build the flow:
`RTSP -> Algorithm -> LLM -> Alarm.`
3. Open the algorithm node and enable LLM review.
4. Fill in the required prompt and options shown in the page.
5. Open the LLM node and confirm the basic settings.
6. Save and run the task.
7. Check the alert list to confirm the result.

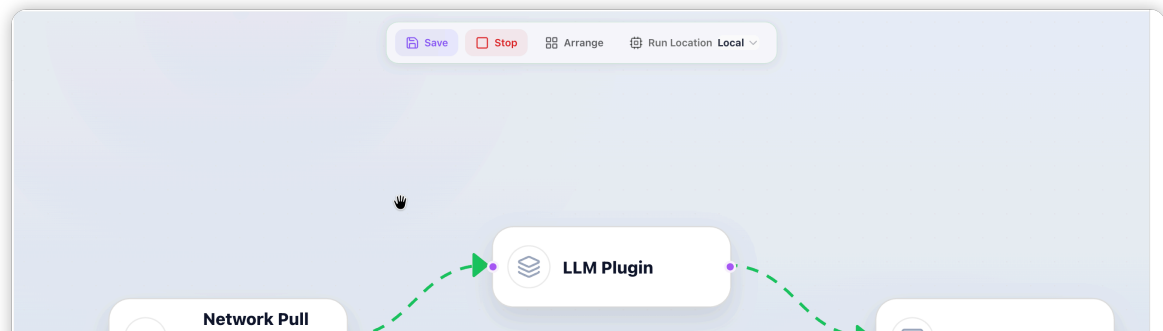




15.3 Detector Mode Steps

1. Create a task and build the flow:
`RTSP -> LLM -> Alarm.`
2. Enable detector mode in the LLM node.
3. Fill in the prompt and interval shown by the page.
4. Save and run the task.
5. Check the alert panel to confirm alarms are generated normally.





15.4 Common Issues

Symptom	Action
LLM configuration has no visible effect	Check node connection and whether the feature is enabled
Results are unstable	Simplify and adjust the prompt
No alarms are generated	Check the task chain, task status, and prompt content

16. Intelligent Retrieval Usage

This chapter only explains how customers can use Intelligent Retrieval in the web UI.

16.1 Function Overview

Intelligent Retrieval supports:

1. Searching images or videos with text.
2. Searching similar images or videos with an uploaded image.
3. Choosing search scope: image, video, or all.

16.2 Before Use

1. The task should already be running normally.
2. Alert snapshots or recordings should already exist.
3. Intelligent Retrieval should already be enabled in system settings.

16.3 Steps

16.3.1 Text Search

1. Open the Intelligent Retrieval page.
2. Enter the search text.
3. Select the search scope.
4. Click search and review the results.

16.3.2 Image Search

1. Upload a reference image.
2. Select the search scope.
3. Click search and review the results.

16.3.3 View Results

1. Click a result item to view details.
2. If it is a video result, jump to the corresponding time point.
3. Use Recording Management for further playback if needed.

16.4 Common Issues

Symptom	Action
No search results	Confirm the task is running and data has already been generated
Too few results	Change the keywords or switch to <code>all</code> scope
Video results look similar	Open the original video and check by time point

17. Settings & System Management

17.1 Open Settings

- Click **gear icon** in top bar

17.2 Menu Items

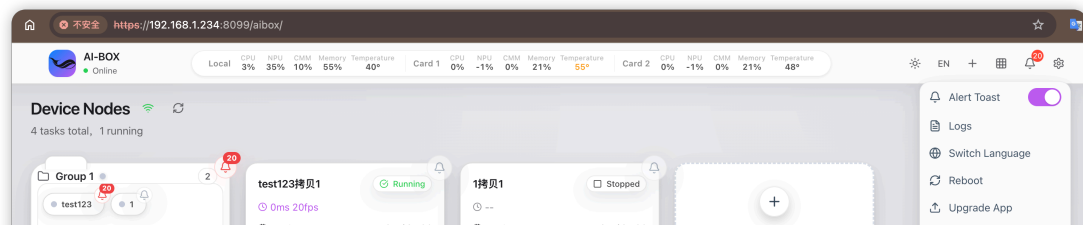
Item	Description
Alert Popup	Toggle alert floating notification
Logs	View, delete device logs
Language	Chinese ↔ English
Reboot	Reboot device (with confirm)
Upgrade App	Upload .deb
Upgrade Firmware	Upload .swu
Network Config	Cloud/access key
Intelligent Retrieval	Configure retrieval enable, run location, frame sampling and thresholds
Version Info	Device, app, firmware versions
Change Password	Change login password
Face Library	See 14.1 (when plugin enabled)
Recording	See 14.2 (when plugin enabled)
Logout	Return to login

17.3 Upgrade App / Firmware

1. Select menu item
2. Drag or select file (app: `aibox-<model>_<ver>-<ts>_arm64.deb`; firmware: `*.swu`)
3. Click "Start Upgrade"
4. Wait; device will reboot

17.4 Device Details

- Double-click status capsule to open device details popup



18. Context Menu Operations

18.1 Task Card Right-click

Option	Description
Edit	Open flow editor
Copy	Copy task
Move to X	Move to group
New Group	Create group and move
Delete	Delete task (red, confirm required)

18.2 Group Card Right-click

Option	Description
Rename	Change group name

18.3 Add Task Card Right-click

Option	Description
New Task	Same as click

18.4 Blank Area Right-click

- "New Task" menu; click to create task

18.5 Close Menu

- Click outside menu to close

19. Comprehensive Example

Example: **Pedestrian detection at entrance + alerts + web preview.**

19.1 Scenario

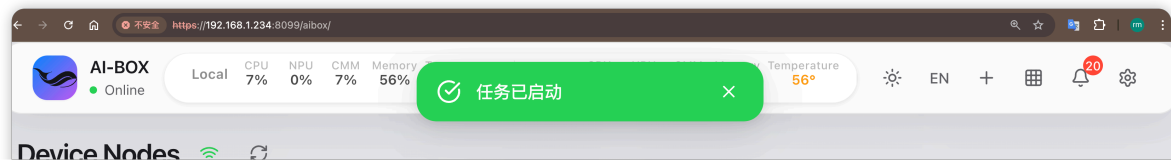
- **Goal:** RTSP camera at entrance, pedestrian detection, alerts on detection, web preview
- **Prerequisite:** Device connected, RTSP URL known (e.g. `rtsp://192.168.1.10:554/stream1`)

19.2 Steps

1. Create task: Add Task → enter name → Create and Edit
2. Add nodes: RTSP Input, Pedestrian Detection, OSD Overlay (optional), Network Output, Alert Output
3. Connect: RTSP → Pedestrian → OSD → Network; Pedestrian → Alert
4. Configure RTSP Input: Enter URL, test connection, set decode location
5. Configure Pedestrian Detection: Thresholds; draw detection region if needed
6. Configure Alert Output: Alert threshold (e.g. 0.8)
7. Save & Run
8. Preview and verify alerts

19.3 Variants

Variant	Action
Preview only	Omit Alert Output
Add face recognition	Chain face recognition after pedestrian; configure face library
Add recording	Add Recording Output node
Multi-input	If supported, add multiple inputs



20. Factory Firmware Plugin Inventory & Capabilities

20.1 Factory-delivered Plugin Packages

Plugin Package	Size	Primary Role
<code>alarm.plugin</code>	959K	Alert output and event dispatch
<code>catdog.plugin</code>	11M	Cat/Dog detection algorithm
<code>facedet.plugin</code>	3.0M	Face detection algorithm
<code>facerec.plugin</code>	106M	Face recognition algorithm (face library matching)
<code>firedet.plugin</code>	26M	Fire detection algorithm
<code>gb28181.plugin</code>	1.1M	GB28181 integration output
<code>hdmplugin</code>	464K	HDMI video output
<code>llm.plugin</code>	1.4M	Multimodal LLM review / detector
<code>lpr.plugin</code>	34M	License plate recognition
<code>motor.plugin</code>	12M	Motor/non-motor related detection
<code>netclient.plugin</code>	693K	Network pull-stream input (RTSP, etc.)
<code>netserver.plugin</code>	466K	Network streaming output
<code>osd.plugin</code>	547K	OSD overlay processing
<code>p2p.plugin</code>	1.3M	P2P access/output
<code>personAttr.plugin</code>	37M	Person attribute analysis
<code>persondet.plugin</code>	37M	Person detection algorithm
<code>record.plugin</code>	1.2M	Recording output and storage

20.2 Detailed Capabilities by Plugin

20.2.1 Input Plugins

1. `netclient.plugin`
 - o Function: pulls camera/network streams (typically RTSP), decodes frames, and feeds the graph.
 - o Typical use: first node in most pipelines.
 - o Typical chain: `RTSP -> Detection/LLM -> OSD/Alert/Stream/Record`.

20.2.2 Algorithm Plugins

2. `persondet.plugin`
 - o Function: detects people and outputs person target metadata.
 - o Typical use: perimeter/entrance pedestrian monitoring.
 - o Can be combined with: `osd`, `alarm`, `record`, `llm` (review mode).

3. `firedet.plugin`
 - Function: detects fire-like regions and emits fire alarm candidates.
 - Typical use: fire safety scenarios.
 - Can be combined with: `llm` review to reduce false positives, then `alarm`.
4. `facedet.plugin`
 - Function: detects face boxes and face quality cues for downstream matching.
 - Typical use: prerequisite stage for face recognition.
 - Typical chain: `netclient -> facedet -> facerec -> alarm/osd`.
5. `facerec.plugin`
 - Function: compares detected faces with configured face library and returns identity/match results.
 - Typical use: whitelist/blacklist/VIP face events.
 - Depends on: valid face library management and `facedet` outputs.
6. `personAttr.plugin`
 - Function: infers person attributes (for example clothing/appearance metadata, model-dependent).
 - Typical use: post-event search and structured description.
 - Typical chain: `persondet -> personAttr -> alarm/record`.
7. `catdog.plugin`
 - Function: detects cat/dog targets and outputs corresponding object results.
 - Typical use: pet monitoring or scene filtering.
 - Typical chain: `netclient -> catdog -> osd/alarm`.
8. `motor.plugin`
 - Function: vehicle/non-motor related detection or categorization (project/model dependent).
 - Typical use: traffic/roadside behavior analysis.
 - Typical chain: `netclient -> motor -> alarm/osd/record`.
9. `lpr.plugin`
 - Function: detects and recognizes vehicle license plates.
 - Typical use: vehicle access control and event indexing.
 - Typical chain: `netclient -> lpr -> alarm/record/server push`.
10. `llm.plugin`
 - Function: used for review mode and detector mode.
 - Typical use: tasks that need a second check or direct LLM detection.

20.2.3 Processing and Output Plugins

11. `osd.plugin`
 - Function: overlays boxes/text/time/status onto video frames.

- Typical use: visualization for preview and streaming output.
- 12. `alarm.plugin`
 - Function: unified alarm handling, snapshot capture, local message insertion, and push dispatch.
 - Supports: local push message, server push message, relay, and TTS playback integration.
- 13. `netserver.plugin` (Network Streaming Output)
 - Function: re-encodes and outputs processed video for live network preview/streaming.
 - Typical use: publish OSD-overlaid stream to web preview or downstream consumers.
 - Typical chain: `... -> osd -> netserver`.
- 14. `hdmi.plugin` (HDMI Output)
 - Function: outputs processed video to HDMI display.
 - Typical use: local monitor wall / screen display on device side.
 - Typical chain: `... -> osd(optional) -> hdmi`.
- 15. `record.plugin`
 - Function: stores video stream to local recording files with management policy.
 - Typical use: event traceability and historical playback.
 - Typical chain: `... -> osd(optional) -> record`.
- 16. `gb28181.plugin`
 - Function: bridges stream/event data to GB28181 platforms.
 - Typical use: integration with standard surveillance platforms in public-sector projects.
 - Typical chain: `... -> gb28181`.
- 17. `p2p.plugin` (P2P Output)
 - Function: enables P2P access path for preview/control integration.
 - Typical use: remote connectivity scenarios where direct LAN path is not preferred.
 - Typical chain: `... -> p2p`.

20.3 Notes

- The plugin list in this chapter is based on the actual delivered firmware version.
- Different projects may have different delivered plugin sets.

21. Troubleshooting

21.1 Connection

Symptom	Cause	Action
"Device connection failed"	Device off, network issue	Check device app, network, firewall
"Device not connected"	Connection lost	Reconnect or refresh
Channel timeout	Device busy, network	Retry; restart device app if needed

21.2 Tasks

Symptom	Cause	Action
Cannot start	Not connected, config error	Check connection, flow
Save fails	References deleted plugin	Remove or replace node
No preview	No Network Output, task not started	Add node, start task

21.3 Plugins

Symptom	Cause	Action
Validation/install takes a long time	Large package is still being processed by the system	Keep the page open and wait
Replacement is rejected	Existing plugin is still referenced by tasks	Stop related tasks and remove the dependency first
Plugin shows "Installed" instead of "Loaded"	The package is already installed, but the service has not fully taken effect yet	Click Restart Service and check again
Disable shows "Pending disable" for a long time	Running tasks still reference the plugin	Stop referenced tasks or restart the service
Uninstall is blocked	Tasks still contain nodes from that plugin	Remove or replace those nodes first
Compatibility warning appears	Plugin-declared platform does not fully match the current device	Verify architecture / OS / chip before deployment

21.4 UI

Symptom	Action
Display issues	Refresh, try another browser
Slow loads	Wait, check network/device load

21.5 Language & Theme

- Language: Settings → Switch language
- Theme: Sun/moon icon in top bar

21.6 Help

- Check device logs
- Contact deployer or support

Appendix A: Shortcuts

Action	Shortcut
Delete selected node/edge	Delete
Arrange nodes	Top "Arrange" button
Fit viewport	Canvas controls / wheel
Region draw: Undo last step	Backspace or right-click
Region draw: Delete vertex/region	Delete
Region draw: Cancel	Esc

Appendix B: Glossary

Term	Description
Node	A unit in the flow (input, algorithm, output)
Edge	Data connection between nodes
Group	Folder for multiple tasks
Alert	Notification from algorithm detection
Schedule	Time-based task start/stop
detail_info	Device-pushed node runtime info
form_status	Device-pushed form live status
formList	Plugin-provided form config

Appendix C: Details

C.1 Task Names

- Name required; unique per device
- Copy appends "Copy"; editable

C.2 Component Panel

- Collapses on small screens (< 640px)
- Categories can fold individually

C.3 Canvas Zoom & Pan

- Wheel: zoom
- Drag blank: pan

C.4 Flow Preview (Expanded Card)

- Single-click card expands to show flow preview
- Running: edges colored by status
- Missing plugins: nodes grayed

C.5 Device Capsule Double-click

- Opens device details popup

End of Document

For updates, refer to the actual UI and product documentation.